Mr Dimelow

Drawing Techniques

Topic	What students need to learn:
3 Processes, techniques and specialist tools continued	3.3 – Use of media to convey design decisions, to record to recognised standards, explain and communicate information and ideas using the following methods and techniques: a) Pictorial drawing methods for representing 3D forms – isometric, 2-point perspective b) Working drawings for communicating 2D technical information – 3rd angle orthographic projection, triangulation c) Nets (developments) for communicating information about 3D forms in a 2D format d) Translation between working drawings, pictorial drawings and nets (developments) e) Report writing.

Learning Outcomes

All – students to be able to:

- recognise the difference between Oblique, Isometric, Single and Two Point Perspective sketches/Drawings
- sketch and draw (using drawing tools) ***two*** 3D objects using oblique and isometric projection and Single and Two Point Perspective

MOST – students to be able to:

- sketch and draw (using drawing tools) ***three*** 3D objects using oblique and isometric projection and Single and Two Point Perspective

SOME – students to be able to:

- sketch and draw (using drawing tools) ***four or more*** 3D objects using oblique and isometric projection and Single and Two Point Perspective

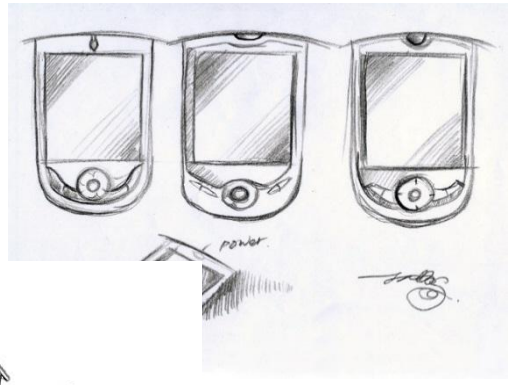
Introduction

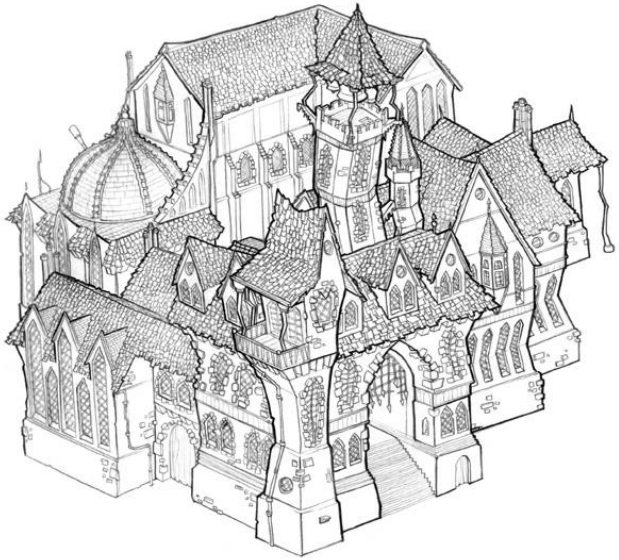
- Graphics forms the basis of communication in all technology areas. It is important that you can use a range of drawing techniques in order to communicate your ideas to other people.
- Drawings can give detailed information about structure, dimensions, materials, and assembly instructions and can be produced using one of a number of formal techniques. The main ones are **oblique projection**, **isometric projection** and - for working drawings - **orthographic projection.**
- It is also very important that you think about the presentation and quality of your work.

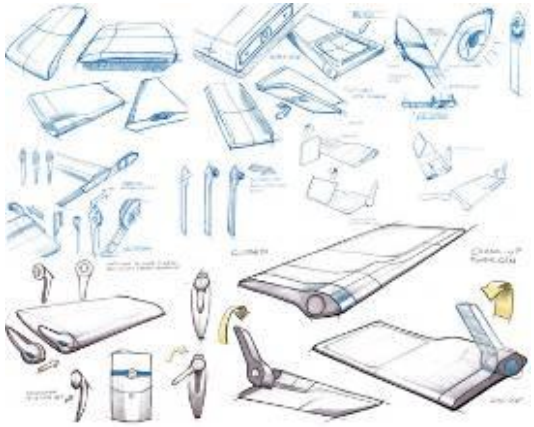
Freehand sketching is very important skill in engineering design.

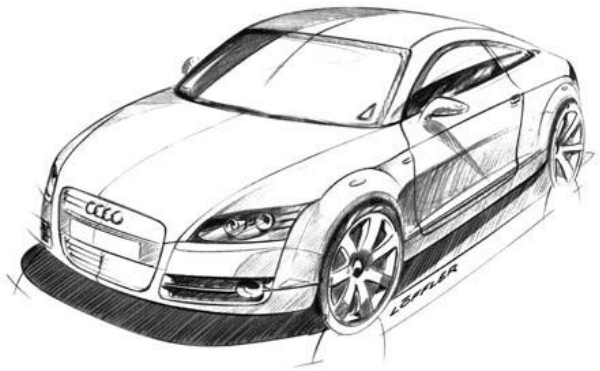
It is very useful for getting your thoughts down on paper quickly.

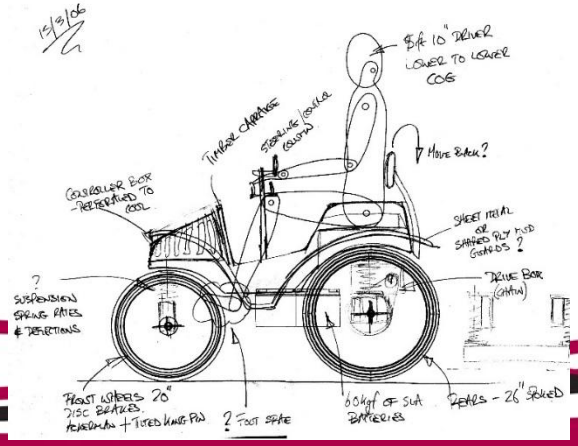
You can develop and communicate your ideas far more easily with sketches than with words.











What is Sketching???

Sketching



Drawing

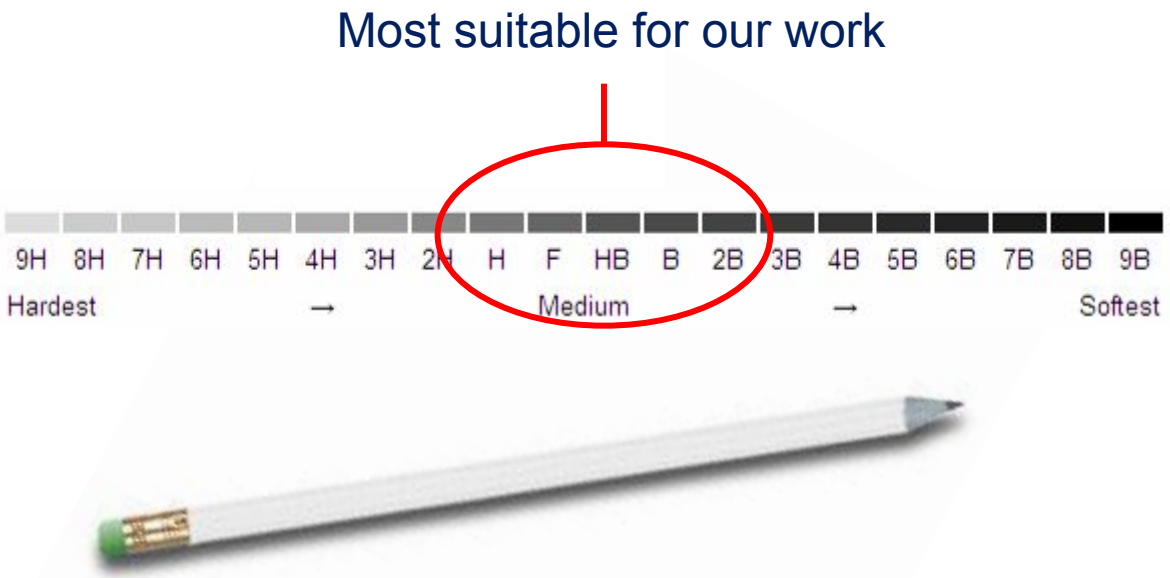


Pencil Grades

The pencil is the most common piece of drawing equipment.

Pencils are available in different grades ranging from the very hard **9H** to the extremely soft **9B**.

2B to **2H** grade pencils are the most suitable for our work.



Free Hand Sketching

Draw a series of horizontal lines and aim to keep them as straight as possible without using a ruler.

- Try to use your whole arm and not just your wrist.
- Hold the pencil lightly but move quickly and freely.

Do about 20 lines, and then try 20 vertical lines.

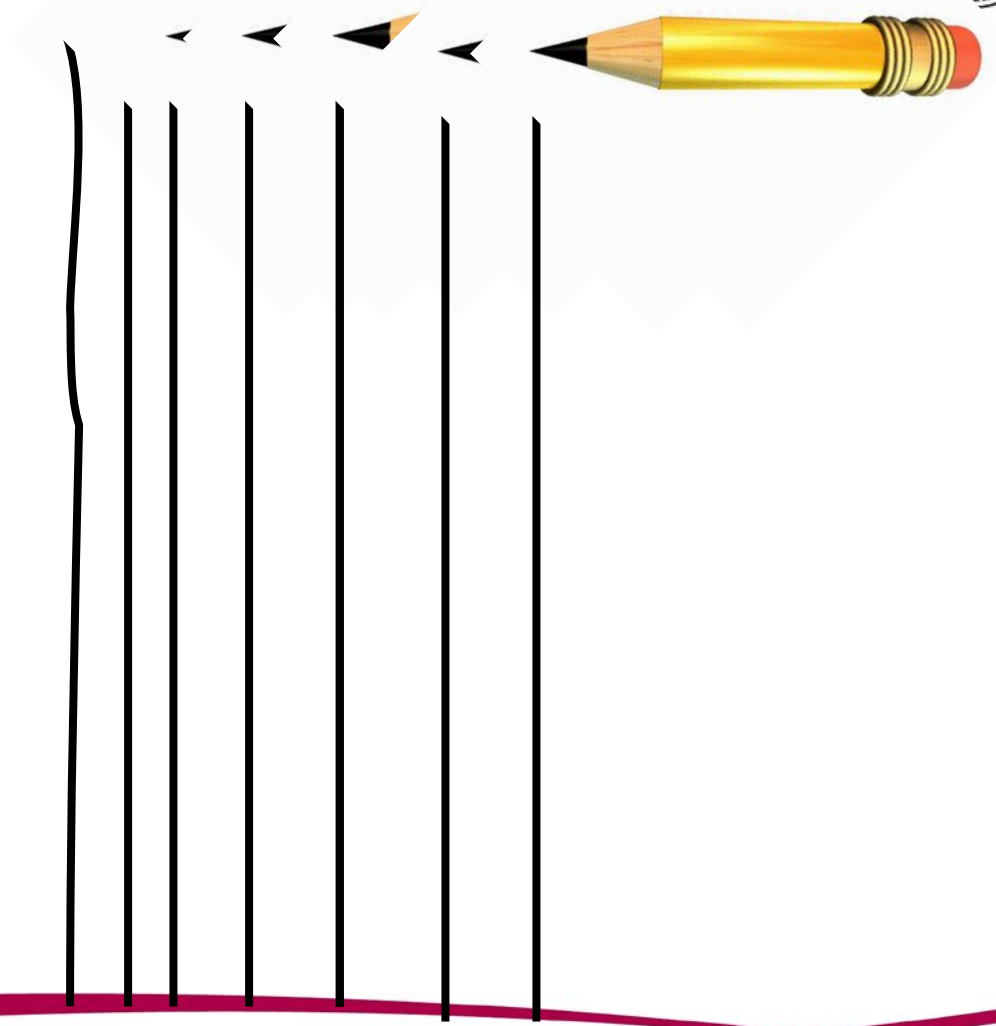


Free Hand Sketching

Draw a series of horizontal lines and aim to keep them as straight as possible without using a ruler.

- Try to use your whole arm and not just your wrist.
- Hold the pencil lightly but move quickly and freely.

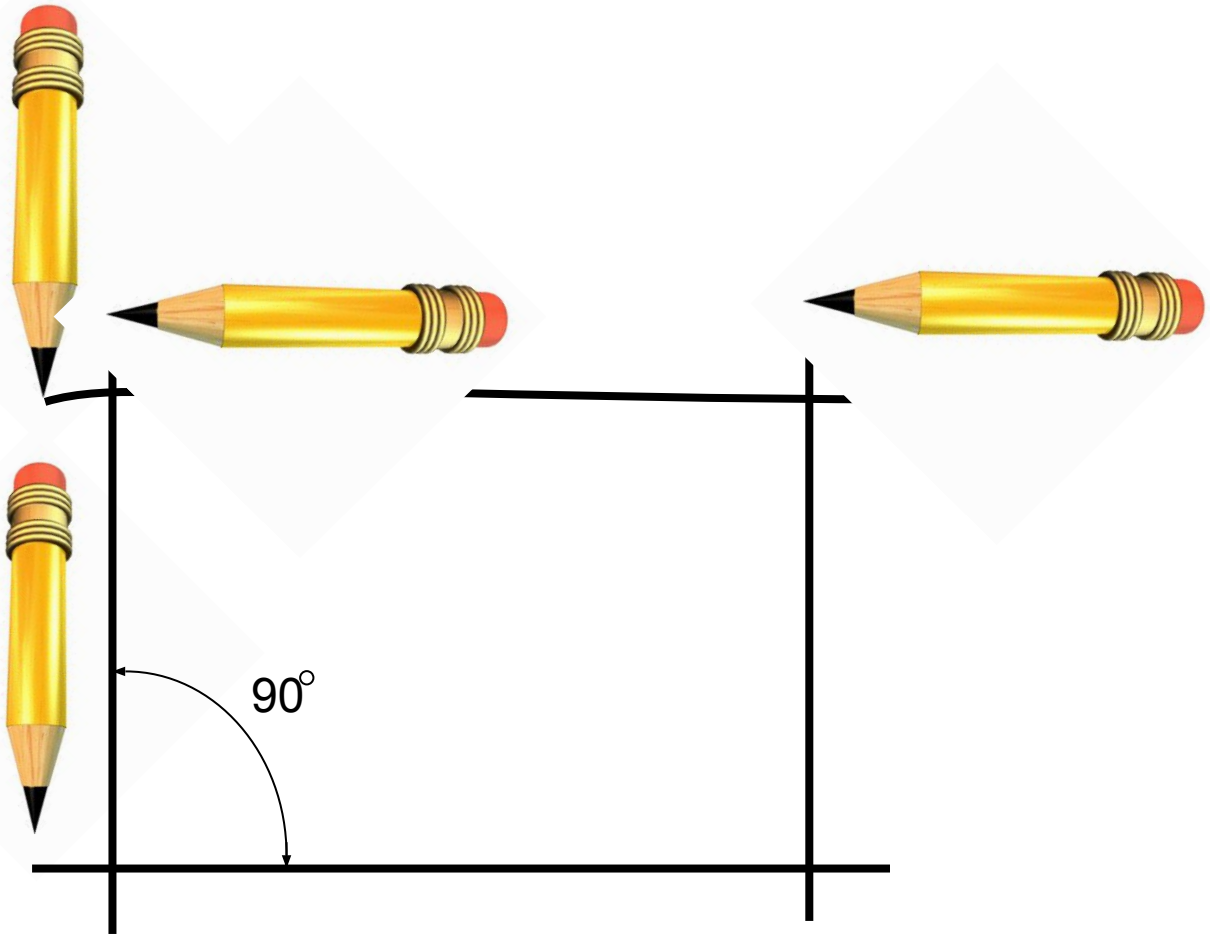
Do about 20 lines, and then try 20 vertical lines.



Free Hand Sketching

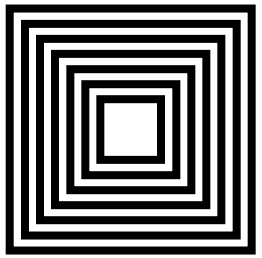
Once you feel happy with pencil control you can practice drawing lines at right-angle.

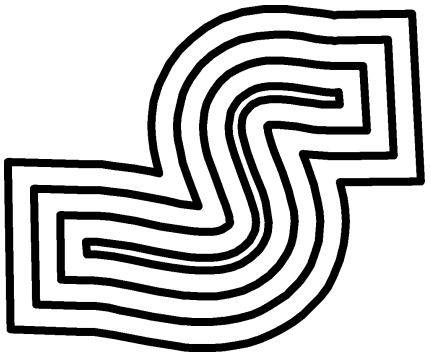
After right-angles, try squares and rectangles.

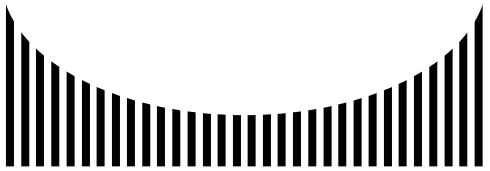


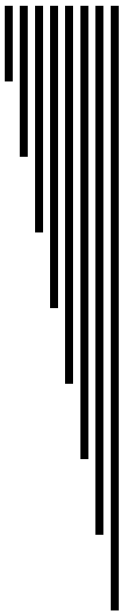
Free Hand Sketching

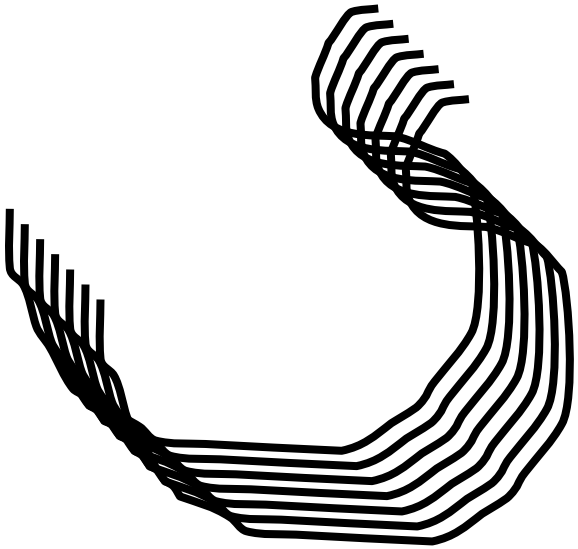
Get even more comfortable with pencil control by trying different shapes and experiments with lines.



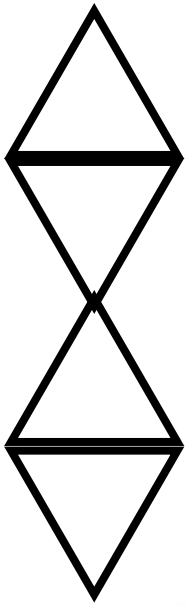








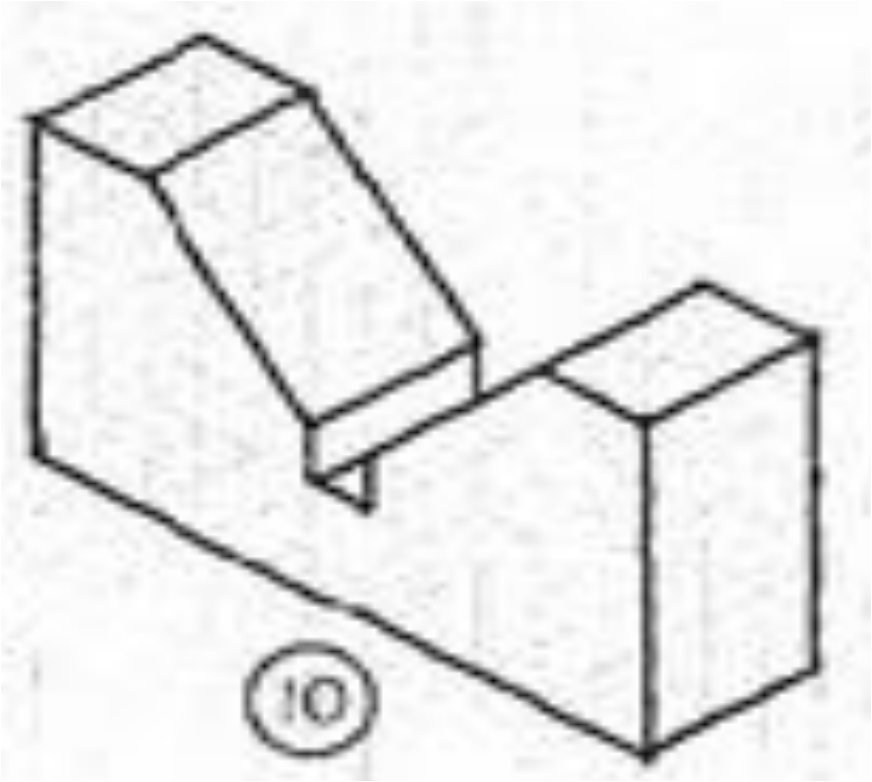




3D Sketching

Quick Task:

Using a sharp pencil quickly sketch the object below in 3D.



Oblique Projection

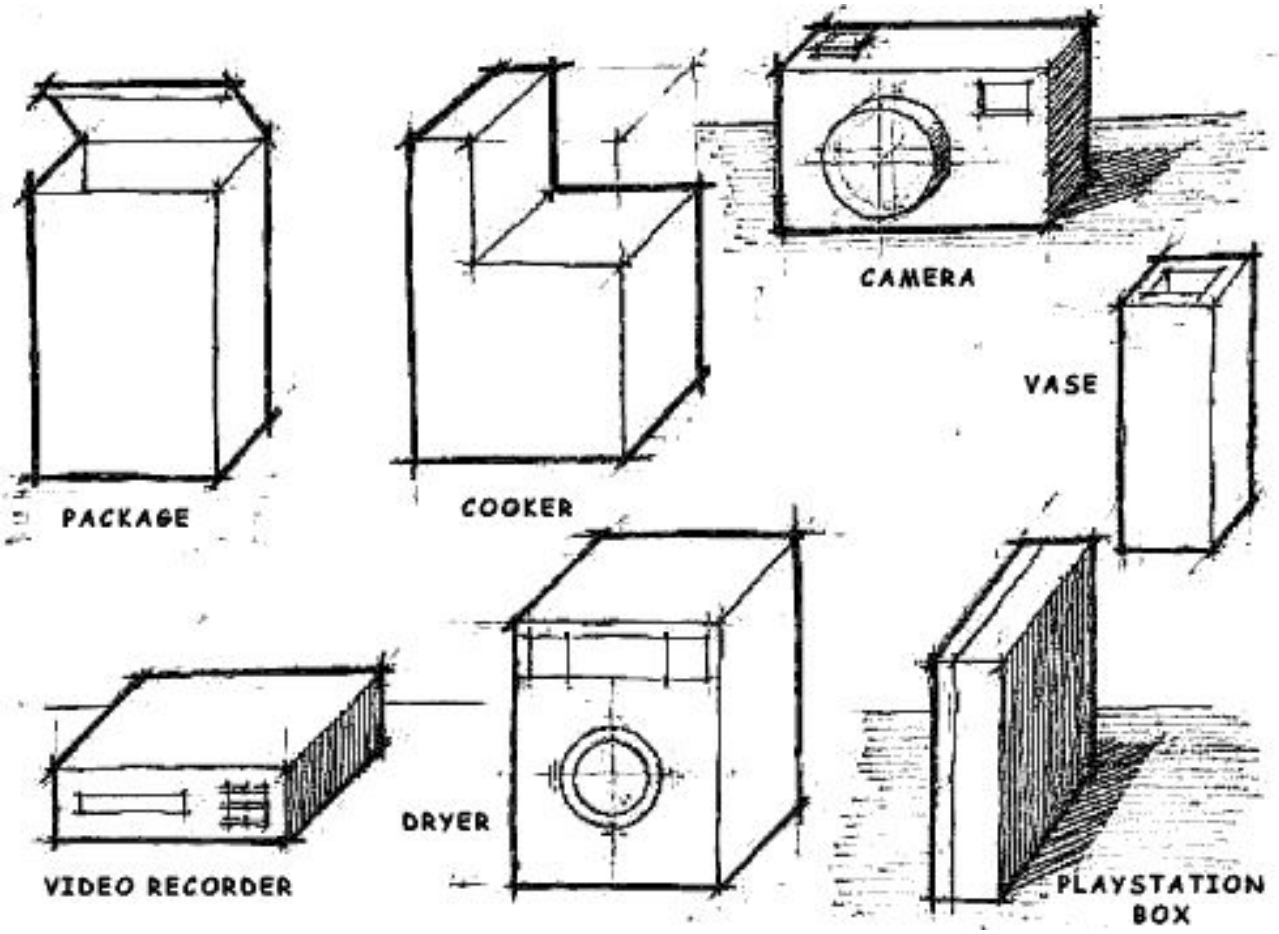
Oblique Projection

- Oblique projection is a method of drawing objects in 3D. The front face of an object is drawn flat on and the oblique lines go back at 45° to the horizontal.
- Most of you draw using oblique but do not realise it.
- When drawing in oblique the oblique line measurements are drawn at half scale in order to make the drawing more realistic. (This is called cabinet oblique, if we draw full length it is called cavalier oblique.)

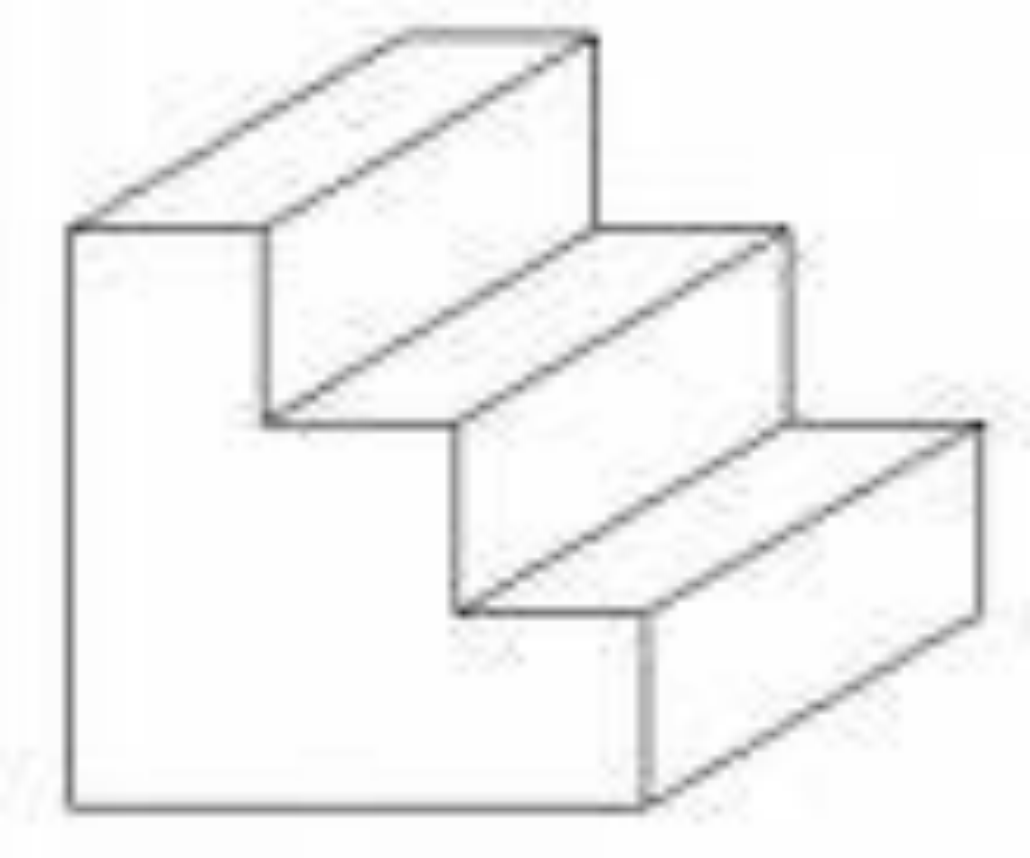
Oblique Sketching Video



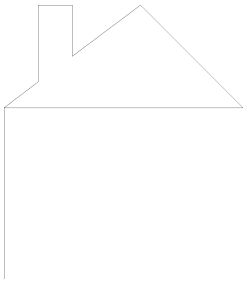
Examples


PACKAGE
COOKER
CAMERA
VASE
VIDEO RECORDER
DRYER
PLAYSTATION BOX

Examples

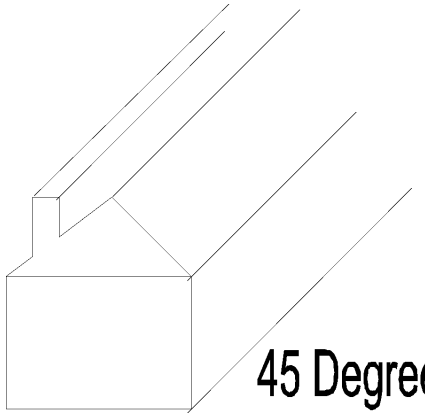


Method

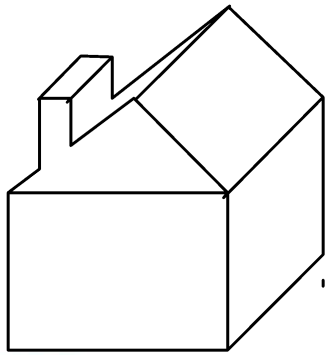


1. Draw the front face of the object flat on (2D)

2. Project lines back from this at 45 degrees



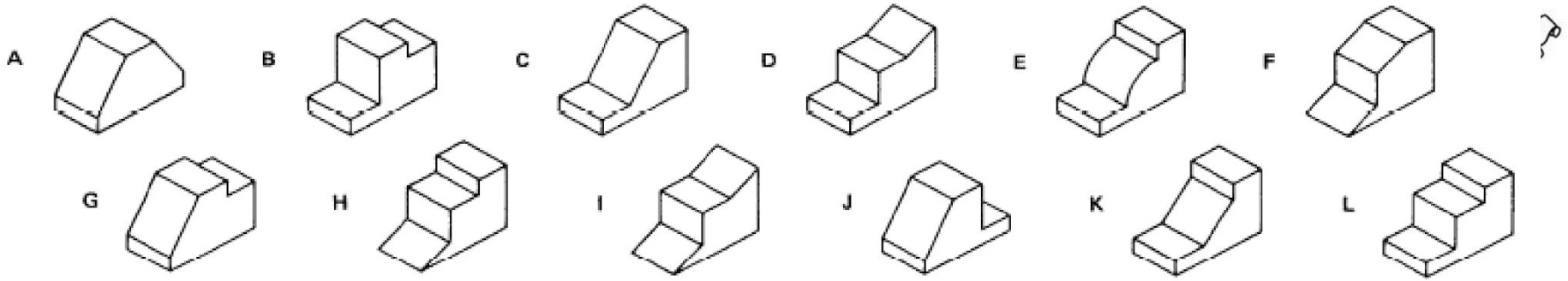
3. Add a depth and complete the outline shape.



4. Leave in construction lines and outline using a fine liner

http://www.technologystudent.com/despro2/obli1.htm

Task



Sketch as many of the shapes above you can in ***10 minutes*** using ***Oblique projection***.

Now use ***drawing tools*** to ***construct*** a minimum of ***two*** of the above shapes using ***Oblique projection***.

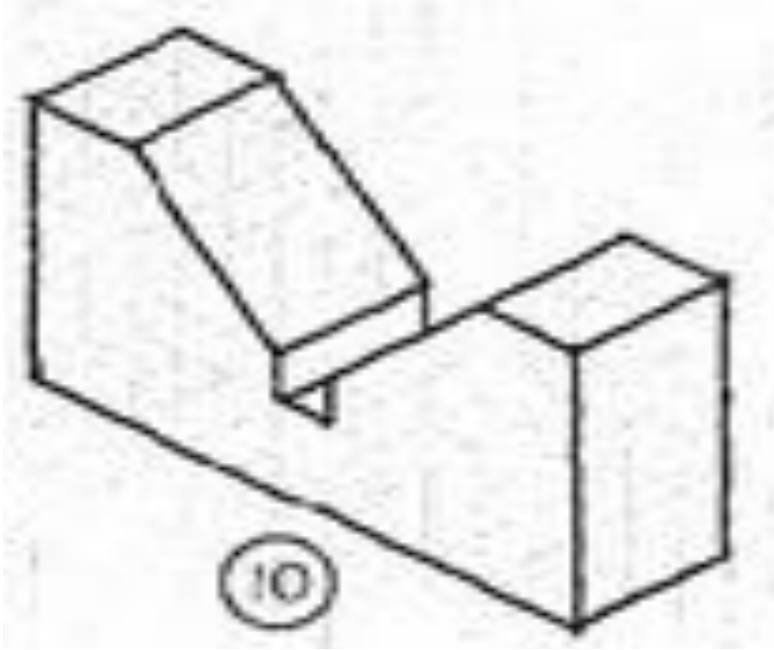
HINT

Remember that all projection lines are drawn half the actual length.

Exit Card:

On the paper provided - Using a sharp pencil quickly ***sketch*** the object below in using ***'Oblique Projection'***.

Make sure you include your ***'name'*** and the ***'date'***.
Hand the work to your teacher.



Isometric Projection

Learning Outcomes

All – students to be able to:

- recognise the difference between Oblique and Isometric sketches/Drawings
- sketch and draw (using drawing tools) ***two*** 3D objects using oblique and isometric projection

MOST – students to be able to:

- sketch and draw (using drawing tools) ***three*** 3D objects using oblique and isometric projection

SOME – students to be able to:

- sketch and draw (using drawing tools) ***four or more*** 3D objects using oblique and isometric projection

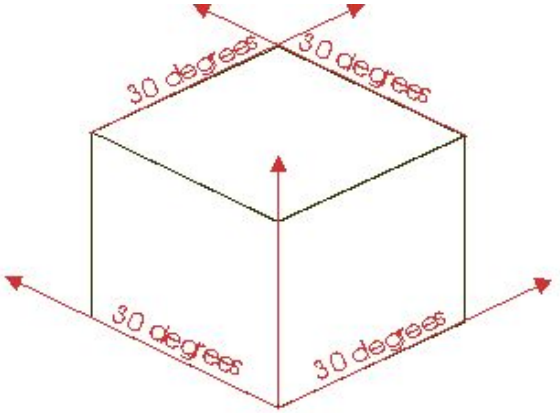
Isometric Projection

Isometric is a method of drawing in three dimensions. It shows three sides of a product on a single view and uses the angle 30 degrees.

It is vital that drawing equipment such as drawing boards and 30/60 degree set squares are used carefully when creating these drawings

Isometric Projection has three main **rules**:

1. All vertical lines stay vertical
2. All horizontal lines become lines at 30 degrees
3. Measurements can be made along these lines



http://www.technologystudent.com/prddes1/drawtec2.html

Isometric Sketching Video

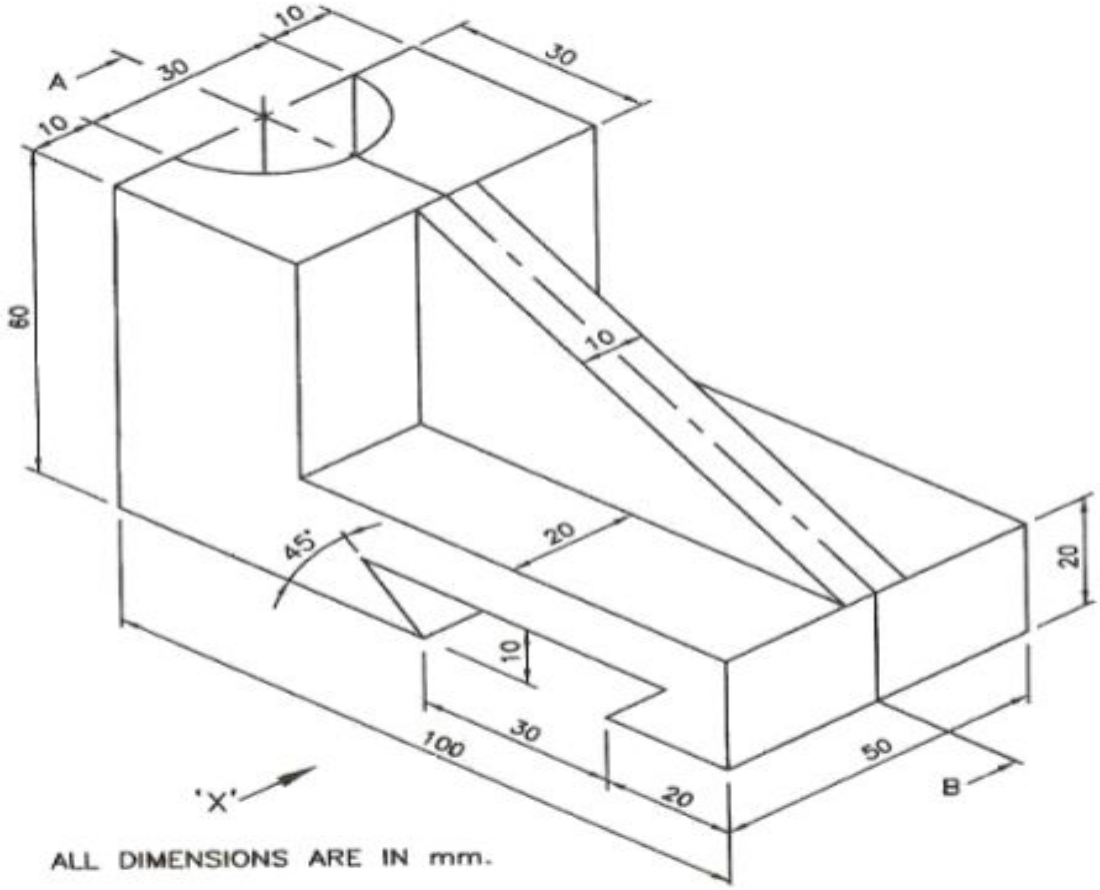


Example

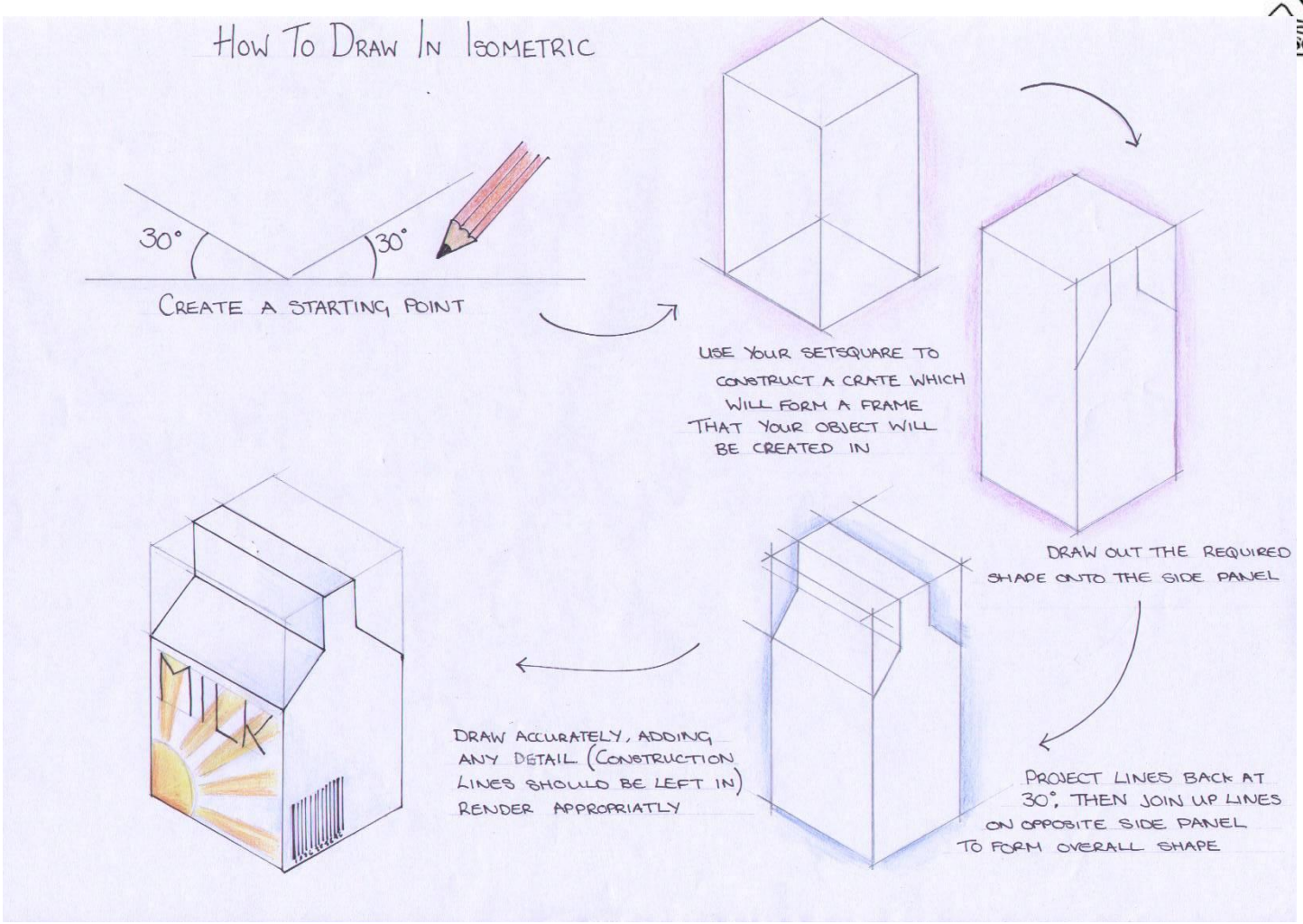




Example


A
10
30
10
30
60
10
45°
20
20
10
30
100
20
50
B
'X'
ALL DIMENSIONS ARE IN mm.

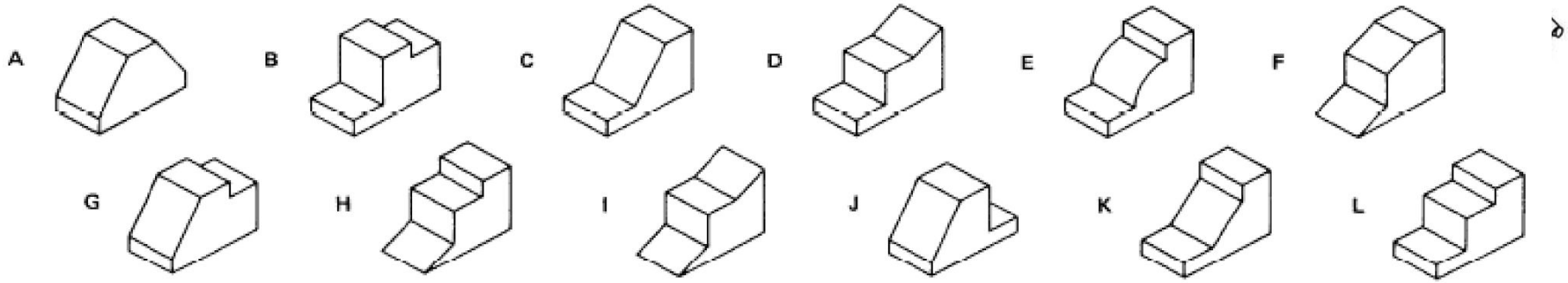
Instructions



Task 1

- Produce an idiots guide to Isometric Projection.
- This could be in the form of a poster or a leaflet.
- DO NOT copy the example shown, choose another object to demonstrate the technique

Task 2



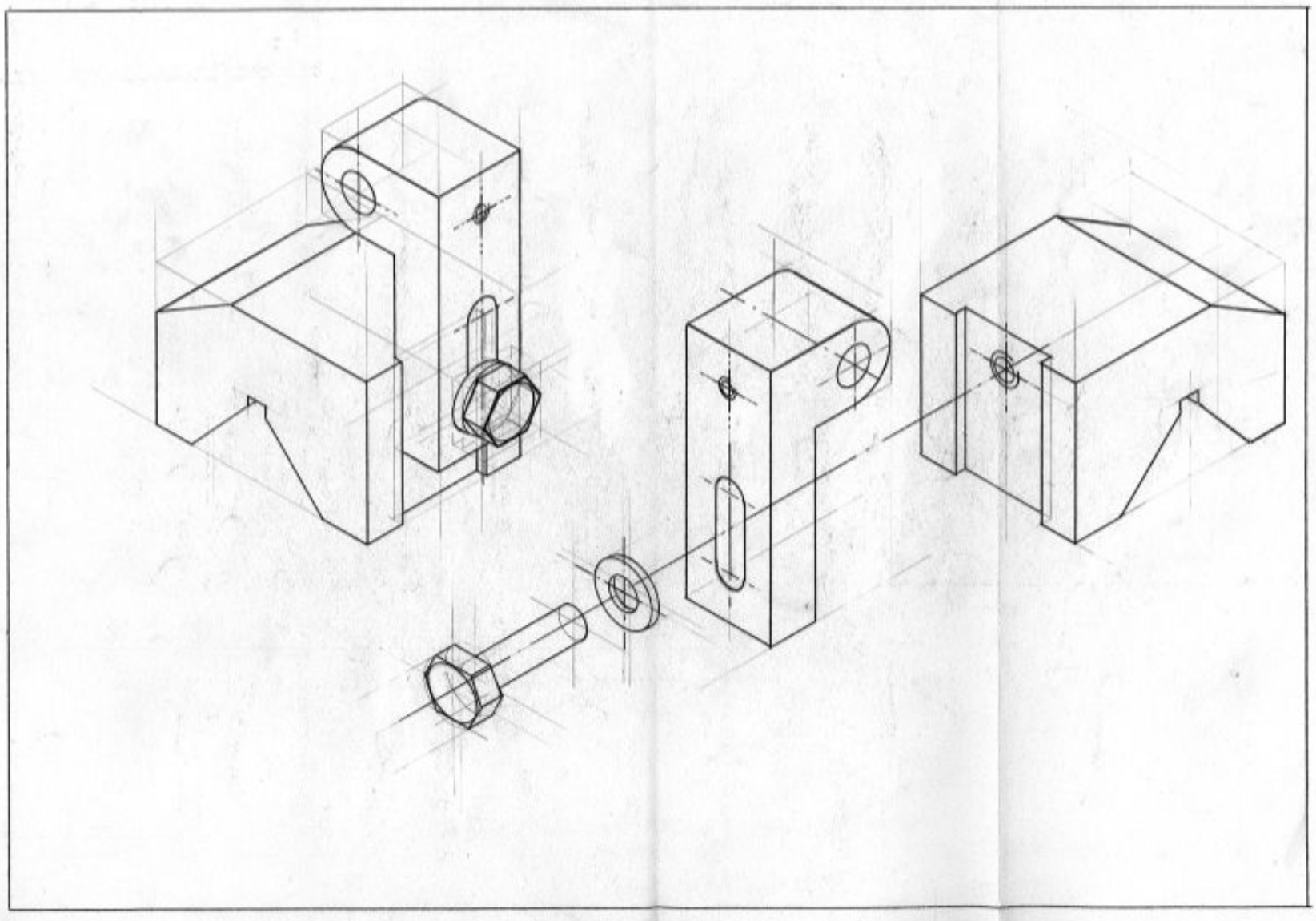
Sketch as many of the shapes above you can in ***10 minutes*** using ***Isometric projection***.

Now use ***drawing tools*** to ***construct*** a minimum of ***two*** of the above shapes using ***Isometric projection***.

HINT

Remember that all horizontal lines are drawn at 30° and all vertical lines are drawn at 90° (vertical).

Examples – Isometric



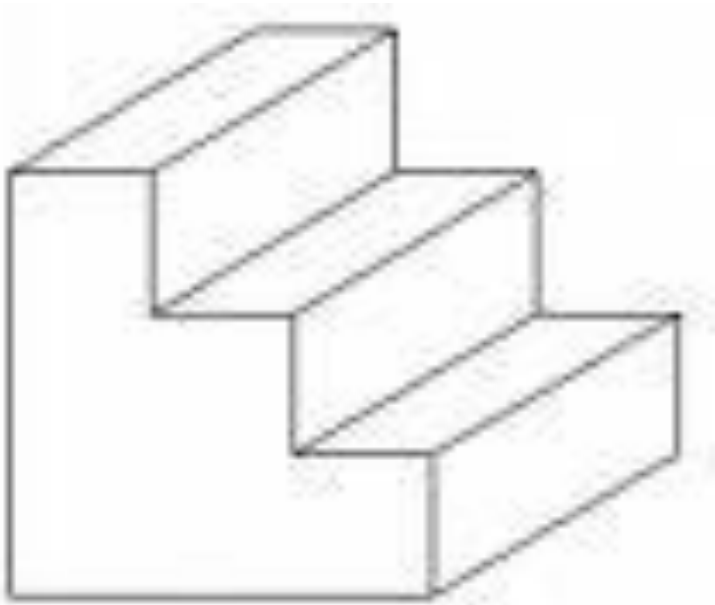
Exit Card:

On the paper provided - Using a sharp pencil quickly ***sketch*** the object below in using ***'Isometric Projection'***.

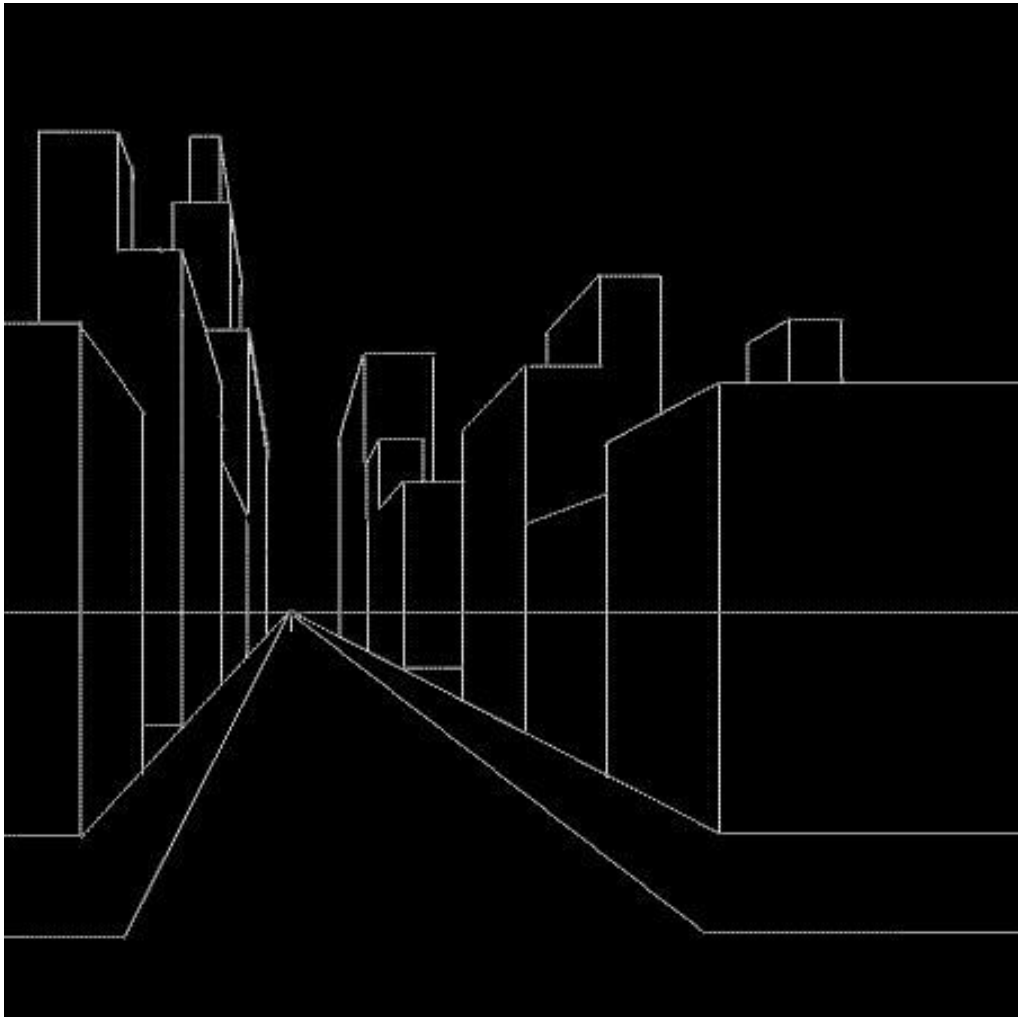
Render (colour) your sketch using ***'three tone rendering'***.

Make sure you include your ***'name'*** and the ***'date'***.

Hand the work to your teacher.



Perspective



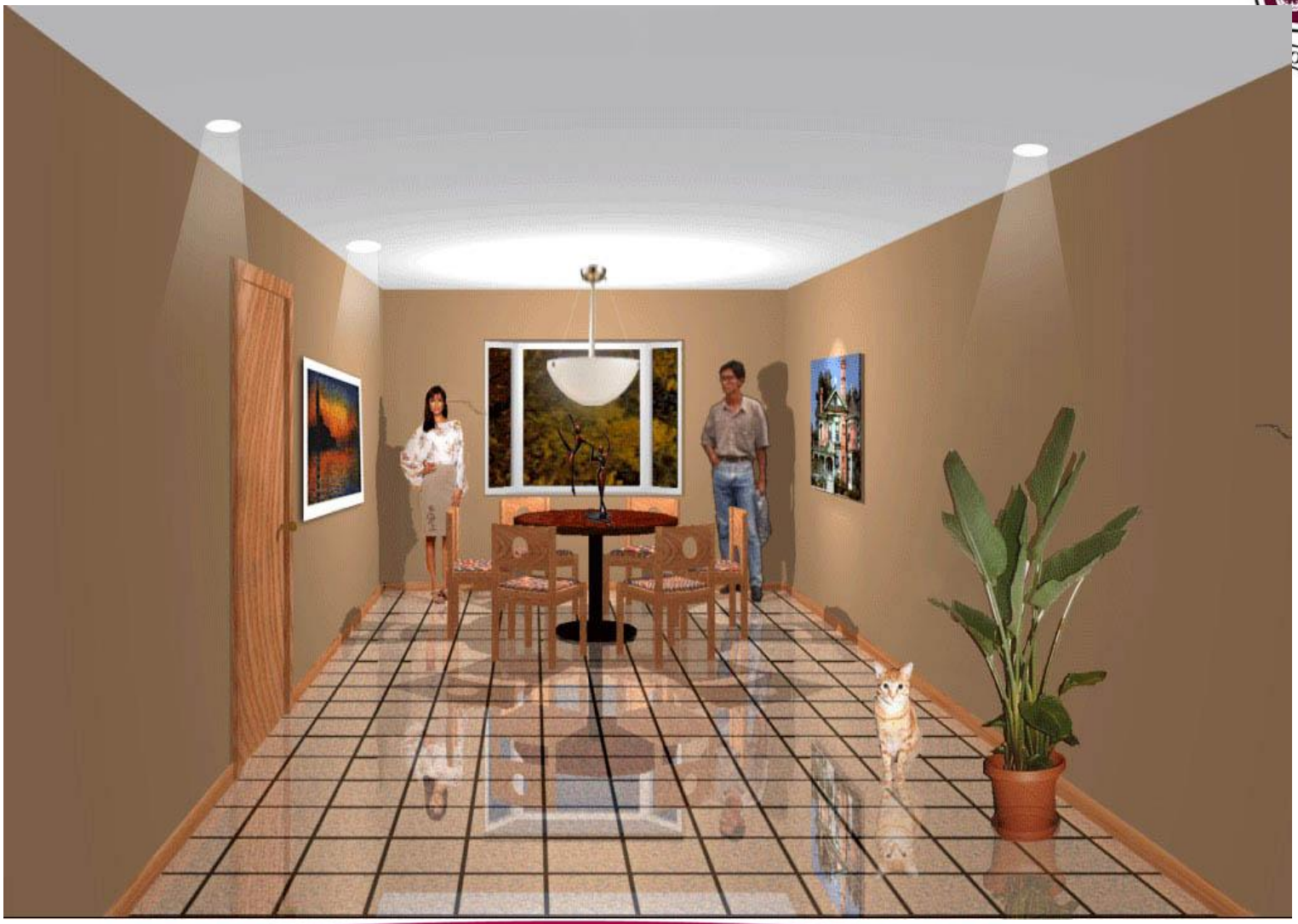
PERSPECTIVE

Perspective drawing can use one, two or three vanishing points. **One-point perspective** is often used for room interiors. **Two-point perspective** has many applications for developing ideas in 3D. **Three-point perspective** is often used for drawings of tall buildings.

The advantage of perspective drawing is that it makes objects appear more realistic, as objects appear to get smaller as their distance from the observer increases.

Perspective drawing can be done using drawing equipment or freehand.

Examples



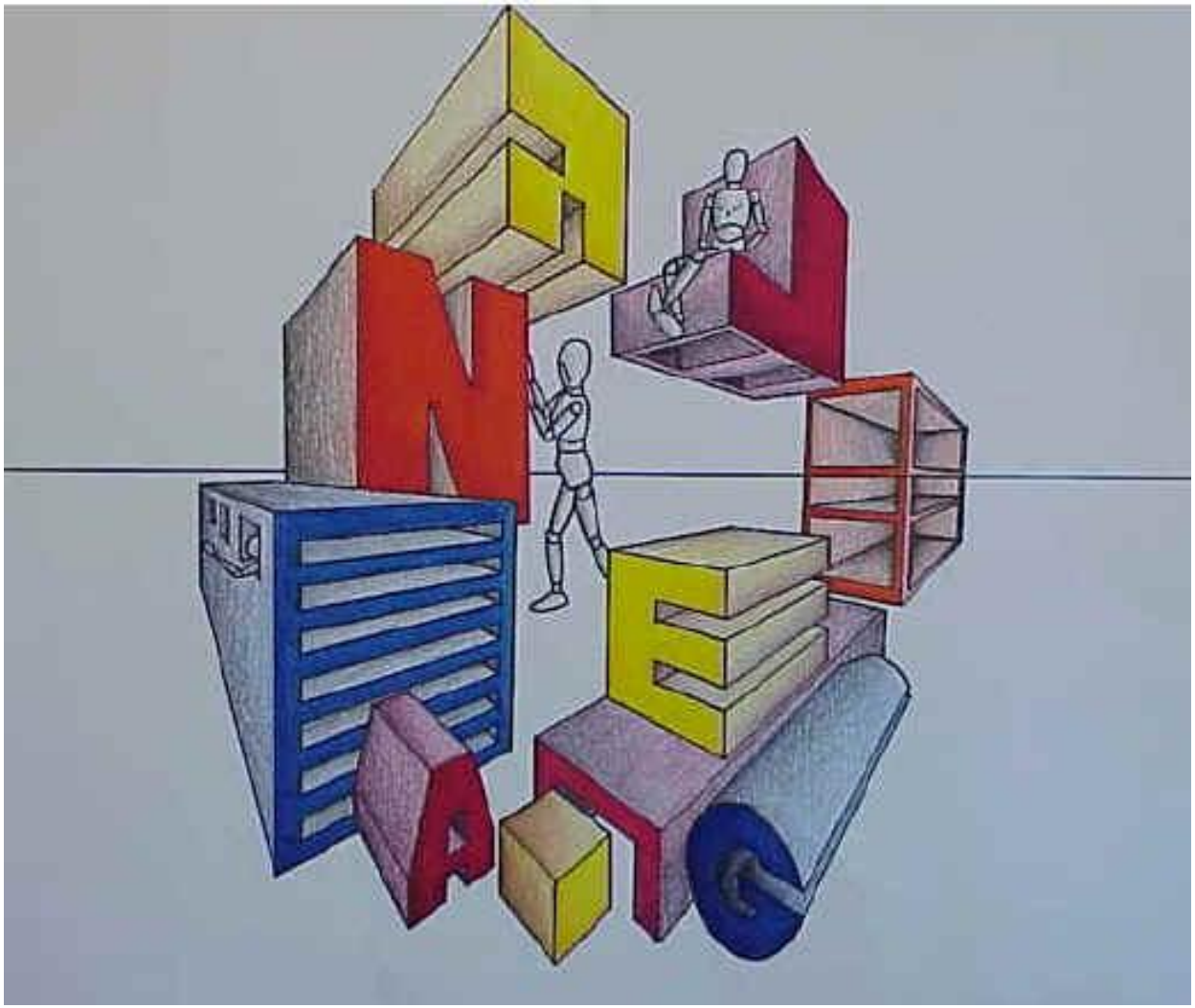
Examples



Examples



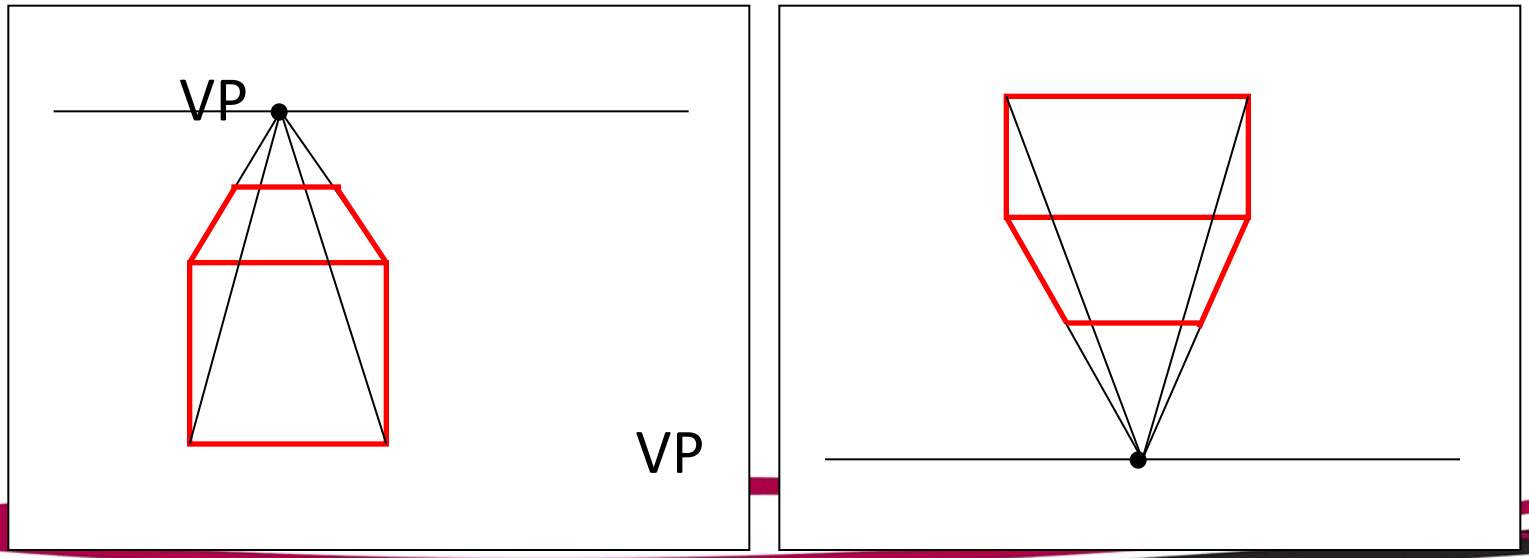
Examples



Single Point Perspective

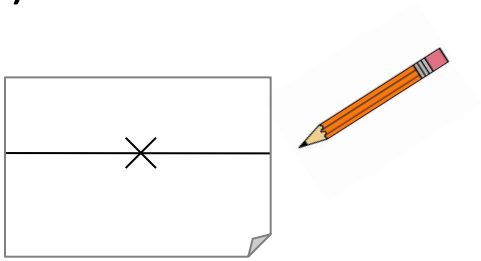
Single Point Perspective

- As the clue in the name says, single point perspective uses just one Vanishing Point.
- The front view of the object is drawn head-on.
- From the object, lines are drawn from the object to the Vanishing Point on the horizon.



Instructions

- Take a ruler and draw one straight horizontal line across the middle of the page. (**Horizon Line**)



- Add a **Vanishing Point** somewhere on that line
- Below the line: draw an accurate square of around 50x50 mm
- From each corner of the square, draw **projection** lines back to the Vanishing Point
- Now add the back of the box, drawing horizontal and vertical lines
- Repeat this process adding approximately 8 **different size boxes** around your page
- Add colour to these boxes by shading them darker at the front and lighter as they go towards the vanishing point

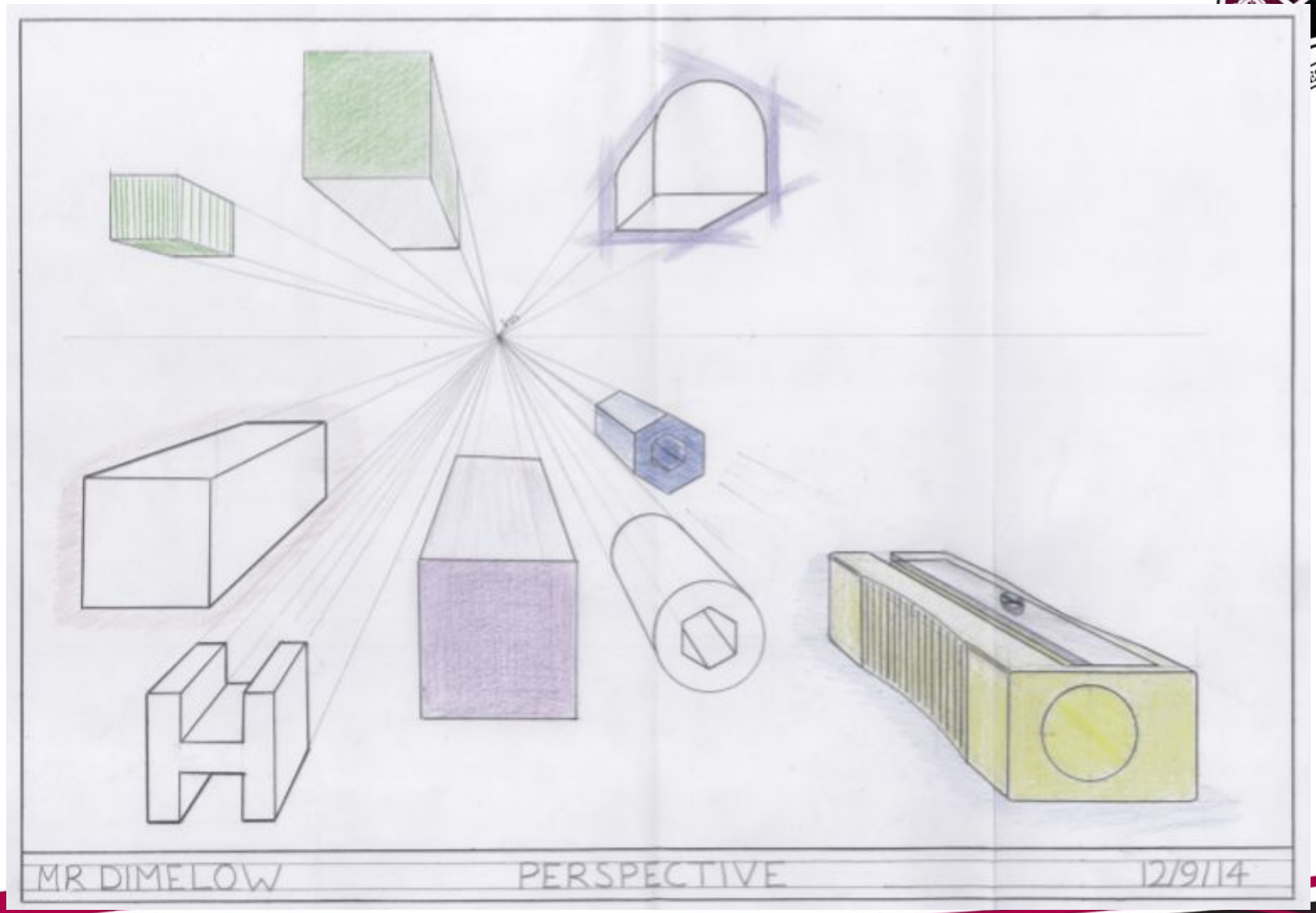
More Complex Items

- Now you understand the basics it is possible to create more complex shapes.
- Take an everyday object such as a pencil sharpener and try to draw it in single point perspective.

HINT

- Start by creating a box
- Think about the **proportion** of the object, you don't want it to look too long and narrow or short and fat
- Use the example sheets as a guide.

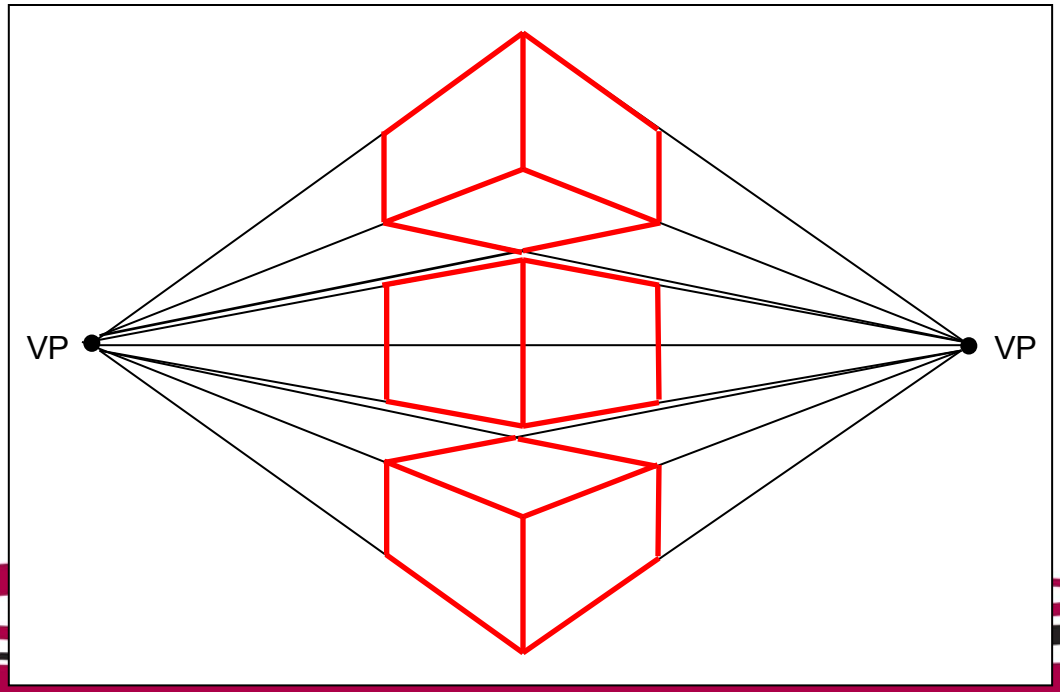
Examples



Two Point Perspective

Two Point Perspective

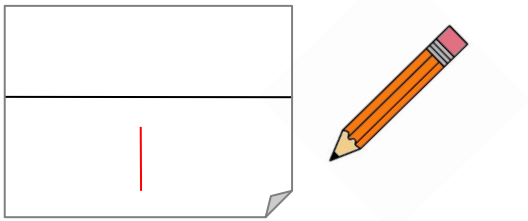
- This type of perspective is more realistic than Single Point perspective.
- This version uses two Vanishing Points.
- Draw the object by starting with a front edge. The ends of the line are projected to the Vanishing Points.



Instructions

1. Take a ruler and draw one straight horizontal line across the page (**Horizon Line**)
2. Add 2 **Vanishing Points**, one at either end of the line.

   Draw one vertical line (50mm). So that it below the **Horizon** line



4. From the top of the line, draw a **projection** line to meet the right hand VP and then the left hand VP. Repeat for the bottom of the line.
5. Give the box two sides by drawing two vertical lines that meet the lines drawn in Step 4.
6. Take these lines back to the opposite vanishing points to complete the box
7. **Repeat** this process to create a **range of boxes and letters** and **render** appropriately.

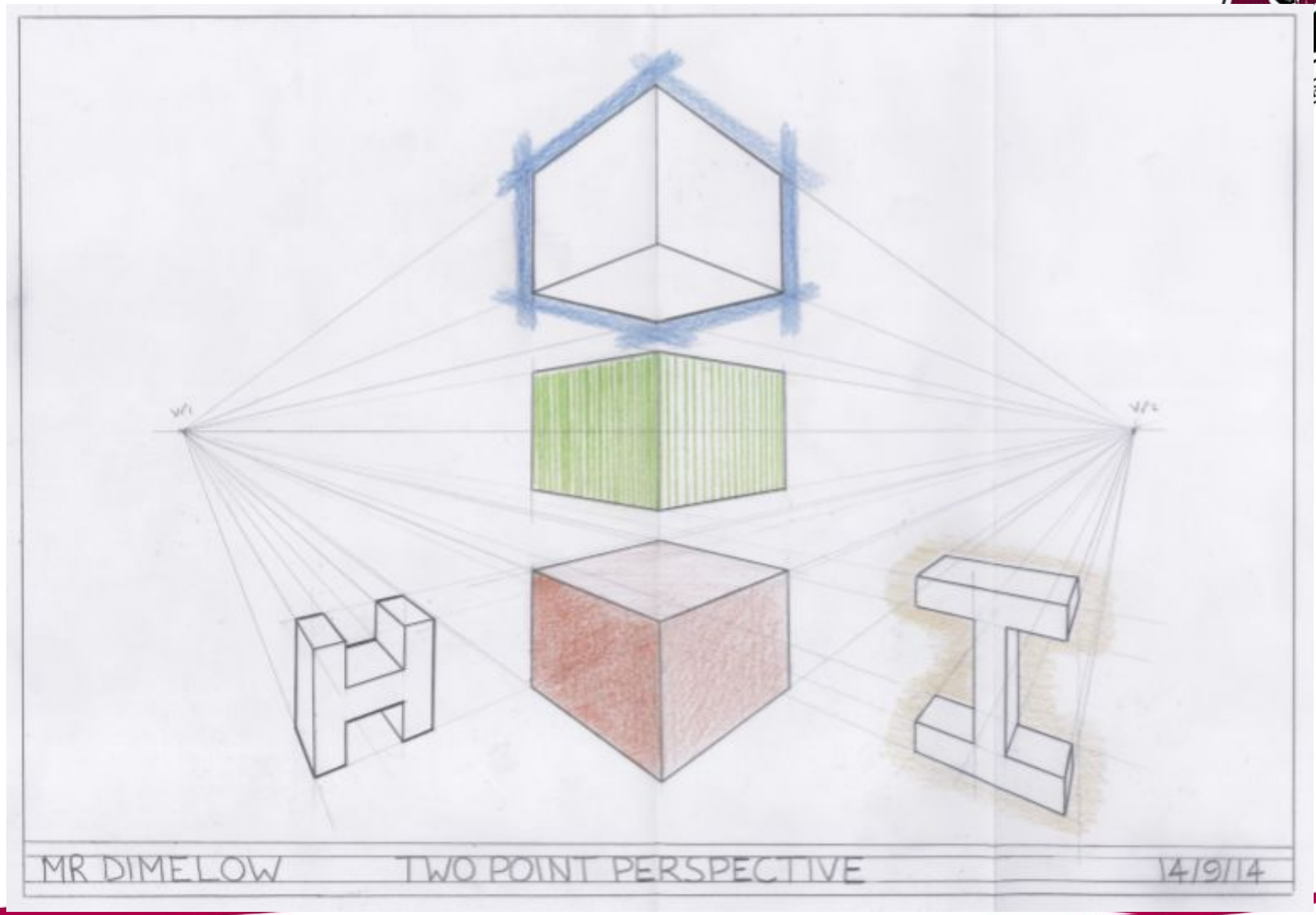
More Complex Items

- It is now possible to use this technique with a variety of objects.
- Take a house and landscape, a street scene, a castle, or another object of your choice and draw it using two point perspective

HINT

- Start by drawing the main edge of your drawing. this may be the corner of your house or castle or the edge of a block of flats in your street scene

Examples



Examples – Isometric + Two Point Perspective

